Digital Image Processing

HW 5: Watershed Segmentation and Object Counting

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Abstract

In medical applications of image processing, often parts of the image need to be segmented and analyzed based on specific aspects such as cells. The issue here is that not all images are perfect and can have each individual cell be separated with ease as many times cells are touching, therefore showing up as a single cell. The watershed method will be used to solve this issue and allow for the total number of cells in an image to be counted automatically.

Introduction

In this paper, images of breast cancer cells and malaria cells will be segmented from their original image and be counted. This will be done through the use of post-processing, distance-transform, thresholding, and watershed. The main goal is to be able to segment cells that could be touching one another, and extract data out of them (count and area).

Experiments and Results

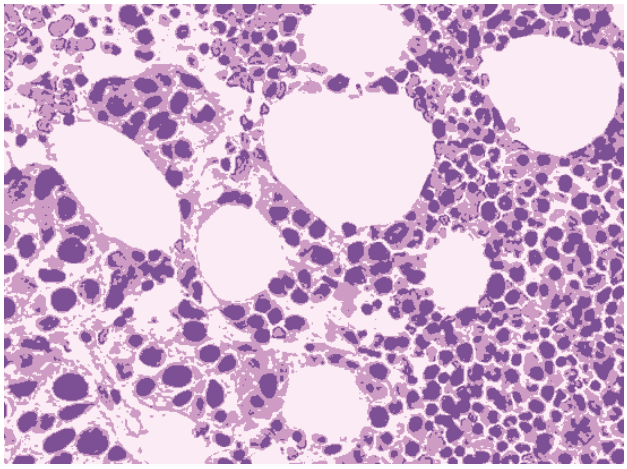


Figure 1. K-means segmentation of breast cancer cells (k=3).

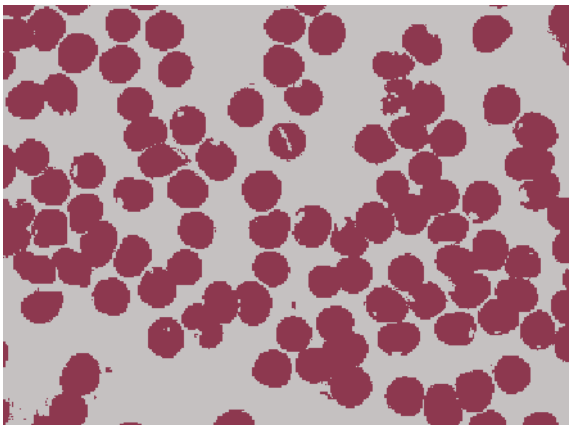


Figure 2. K-means segmentation of malaria cells (k=2).

The first step in this process is to segment the images into clusters based on the colors. Seen in figure 1 and 2, the cell clusters are successfully segmented, with the cells being colored purple and maroon respectively.

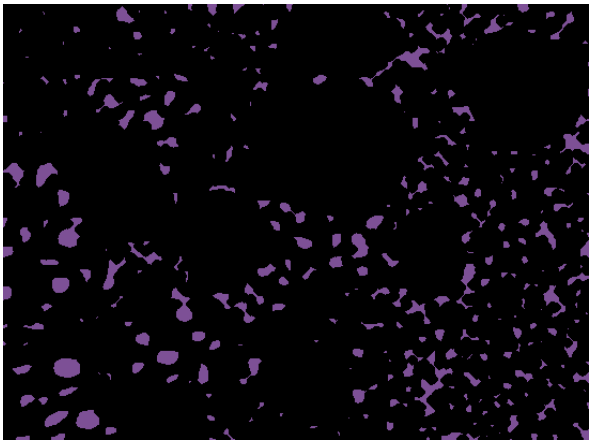


Figure 3. Post-processed mask of the breast cancer cell cluster.

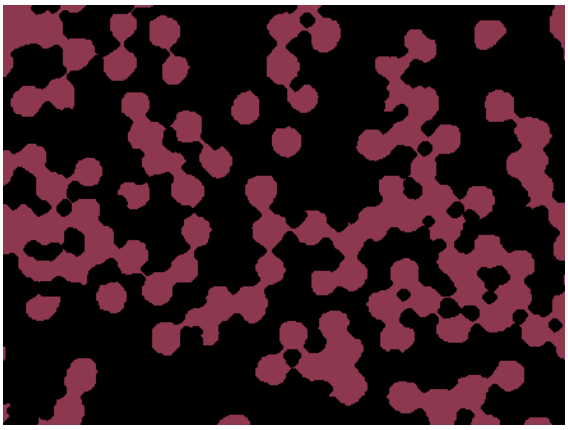


Figure 4. Post-processed mask of the malaria cell cluster.

In order to further prepare this cell mask to be segmented with watershed, morphology transformations were performed for the results of figure 3 and 4. First closing was applied in order to close up the small holes within the cells using a (4,4) ellipse for the structuring element. Then erosion was applied to slightly emphasize the gaps between connecting cells using a (3,3) ellipse for the structuring element.

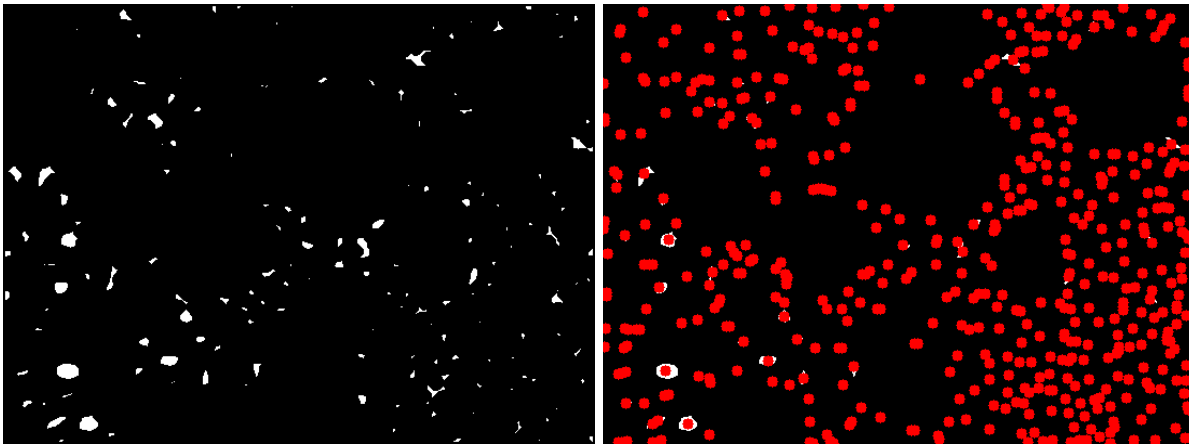


Figure 5. Distance-transform transformation and markers on breast cancer cell masks.

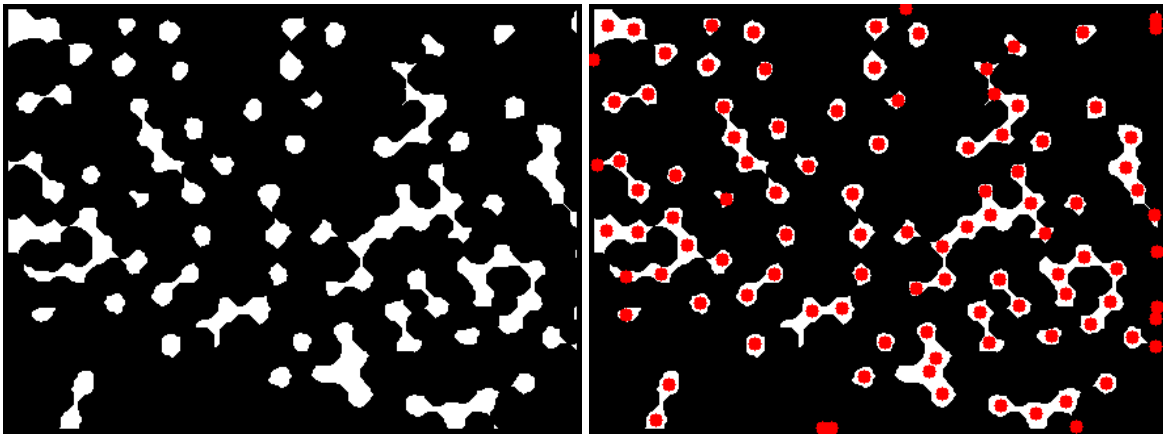


Figure 6. Distance-transform transformation and markers on malaria cell masks.

The distance-transform transformation is applied to the image, which computes the distance from the background to each pixel, basically creating a heatmap of where the closer to the cell center it is, the higher the value is. Using this information, the image was scanned for local maximas, which in turn were used as the markers for each cell center. Each cell center is given a unique label using the connected components function.

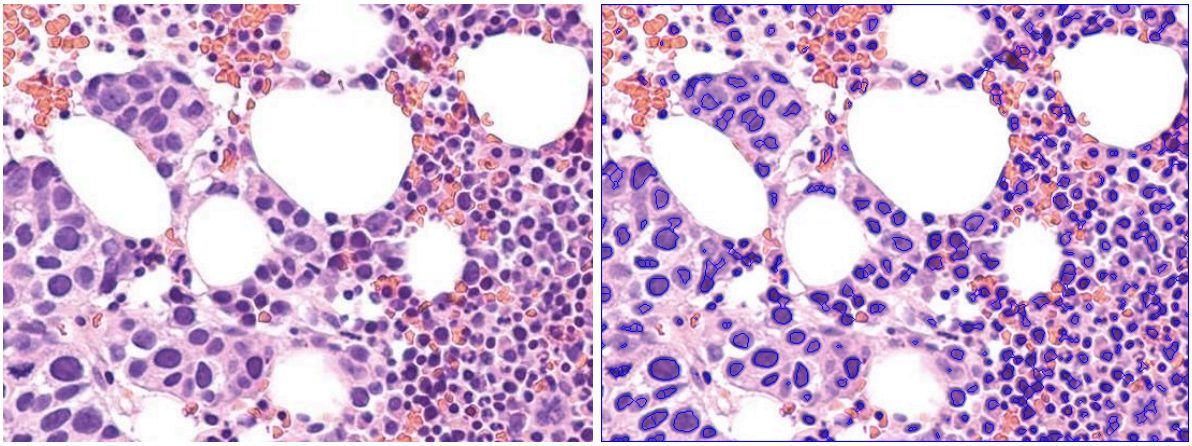


Figure 7. Original and watershed segmented images of breast cancer cells.

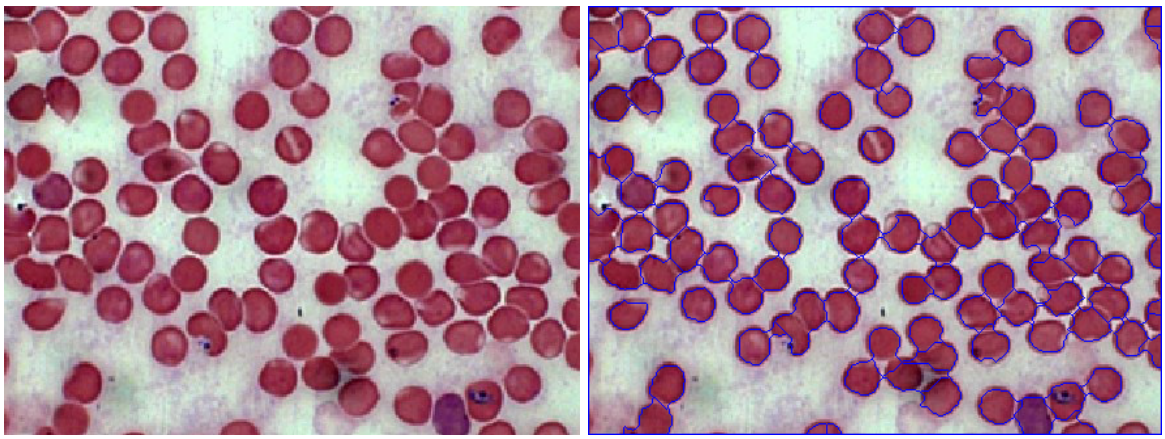


Figure 8. Original and watershed segmented images of malaria cells.

Figure 7 and 8 show the results from the watershed function being used after the previous steps. The distance-transform mask and the markers were used in the watershed process to find where the cells needed to be separated, then an outline was created for all the cells and applied to the original cell images. With this watershed information, the cell count and the total area of cells can be calculated.

Image	**Cell Count**	**Cell Area (Total Pixels)**
Breast Cancer Cells	491	15,664
Malaria Cells	103	50,231

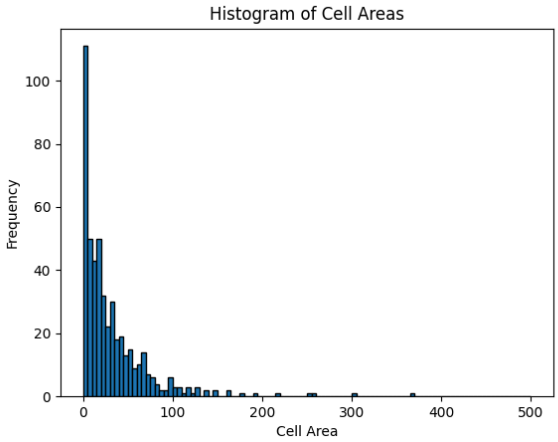


Figure 9. Histogram of the cell areas of the breast cancer cells.

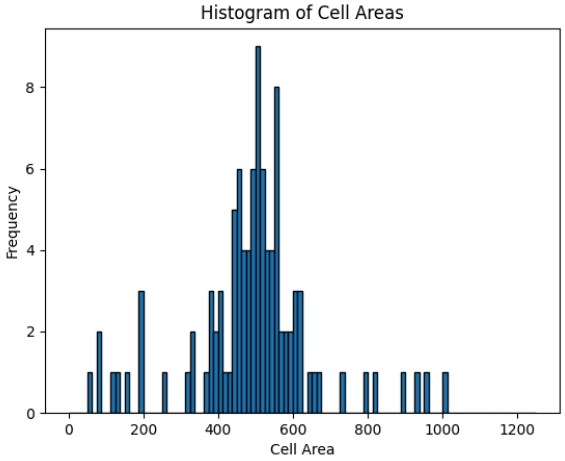


Figure 10. Histogram of the cell areas of the malaria cells.

The histograms in figures 9 and 10 represent the distribution of area sizes of cells within the images. Seen in figure 9, most of the cells in the breast cancer image appear to be smaller. While in figure 10, it shows that the cell size is more biased towards 500 pixel, medium sized range.

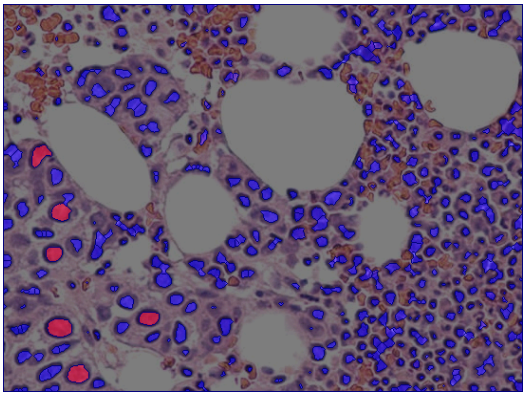


Figure 11. Color watershed mask of breast cancer cells based on size.

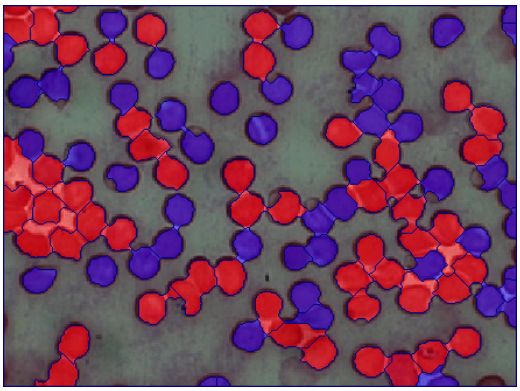


Figure 12. Color watershed mask of malaria cells based on size.

In figures 11 and 12, the cells are categorized into two different colors, red or blue. The colors are decided on by the area of each cell, whether it is below or above the threshold. The threshold is calculated by taking the max sized cell in the image and dividing it by two.

Discussion

The results from the watershed segmentation process were decent, but definitely not perfect. There are a few cells in the images that have mistakes, either aren't separated into two cells as it should, or just aren't being detected at all. The big challenge of this process is finding the best parameters for each image so that each cell is accounted for appropriately. In the main file for this project, the main parameters for the segment function are tweaked for each image for

results in figures 7 and 8, with the parameters being the following: amount of clusters for k-means, the cluster index in which the cells are within, minimum distance between peaks in distance transform, and threshold value for further separating the connected cells. The last two parameters were the most important in changing the end result of the watershed, as depending on the cell size of each image, specific parameters need to be set to prevent cells not disconnecting or cells being discarded completely.

Another issue that occurred during this project was distance-transform wasn't picking up the cells that were on the border of the image. To fix this, the mask image was padded when calculating the markers, then unpadded afterwards, which allowed the cells touching the border to be recognized. This caused another problem of the border being picked up in the area calculation, which was resolved by finding the areas of the cells that are within the range of sizes on the histograms from figure 9 and 10.

Seen in figure 12, cells in which watershed fails a bit (picks up background area when in large cell groups) are emphasized as they are all colored red due to their extra area added by error. Though it doesn't affect the total count output of the image, it does affect the total area of the cells. Additionally, the distributions from figures 9 and 10 can be seen in action through the color mask as well.

The end results using this pipeline were quite successful and majority of each individual cell were segmented from the images, though some adjustments made could potentially improve it more. Configuring with more post-processing and segmentation processes could lead to a more accurate representation of the cells. This would come at a cost of time by potentially being more computationally expensive and adding more parameters that need to be adjusted for each type of

image. This process would be good for using it on similar consecutive images as the parameters wouldn't need to be adjusted, for example if it were just a bunch of samples of the malaria cells, then one set of parameters would be sufficient.

Works Cited

"OpenCV: Image Segmentation with Distance Transform and Watershed Algorithm." *Docs.opencv.org*, docs.opencv.org/3.4/d2/dbd/tutorial_distance_transform.html.

"OpenCV: Morphological Transformations." *Docs.opencv.org*, docs.opencv.org/4.x/d9/d61/tutorial_py_morphological_ops.html.

"Finding Local Maxima — Skimage 0.25.2 Documentation." *Scikit-Image.org*, 2025, scikit-image.org/docs/0.25.x/auto_examples/segmentation/plot_peak_local_max.html.